

JASKEERAT RAI

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SUMMARY

Fourth-year Software Engineering student at the University of Calgary with hands-on experience building full-stack AI, ML, data engineering, and backend systems. Built and deployed production-style applications using Python, FastAPI, React, TypeScript, PostgreSQL, Docker, PyTorch, Airflow, Terraform, and LLM APIs. Experienced in REST API design, normalized schemas, ETL pipelines, model evaluation workflows, and infrastructure as code, with a portfolio spanning AI job matching, audio ML, sports prediction, simulation systems, and data pipelines.

EDUCATION

University of Calgary, Schulich School of Engineering
Bachelor of Science in Software Engineering (ENSF), 4th Year
Coursework: Data Structures and Algorithms, Software Architecture, Databases, Object-Oriented Programming, Operating Systems, Probability and Statistics

Calgary, AB
Expected Graduation: May 2028

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript/JavaScript, C++, Java
AI & LLMs: Claude API, prompt engineering, RAG pipelines, LLM evaluation, agentic workflows, structured output parsing
ML & Data: PyTorch, scikit-learn, pandas, NumPy, feature engineering, model evaluation, signal processing
Backend & Data Engineering: FastAPI, PostgreSQL, SQLAlchemy, Alembic, Apache Airflow, dbt, REST API design, ETL pipeline design
DevOps & Infrastructure: Terraform, AWS (ECS Fargate, RDS, ALB, VPC, IAM, ECR, SSM, CloudWatch), Docker, GitHub Actions CI/CD, Trivy, tfint, Infracost, Git
Frontend & Visualization: React, TypeScript, Tailwind CSS, Recharts, Power BI, Three.js, React Three Fiber

EXPERIENCE

AI Training Specialist | Outlier (Remote)
• Evaluated 1,000+ AI-generated responses for coding, reasoning, SQL, and data-processing tasks against ground-truth criteria, reducing false-positive rates 20% through documented failure-pattern analysis.
• Reviewed Python scripts and SQLite queries to identify and resolve errors and ambiguities, improving reliability and consistency of LLM-assisted evaluation workflows.
• Documented recurring LLM failure patterns, including weak edge-case coverage, incorrect SQL logic, and unsupported factual claims, to improve model evaluation consistency.

Software Developer Intern | Humility Kindness Love (Remote)
• Developed and maintained full-stack internal software systems (FastAPI, PostgreSQL, Tailwind CSS, Docker Compose) improving efficiency and accuracy in event operations, user submissions, admin workflows, and reporting.
• Built REST API endpoints and backend logic that streamlined data management and enabled non-technical stakeholders to submit forms, track events, and generate accurate reports.
• Added structured error handling and input validation to backend APIs, reducing recurring data-consistency issues in production use.

PROJECTS

GapCheck | AI Job-Fit Platform
Claude API, FastAPI, PostgreSQL, SQLAlchemy, Terraform, AWS, GitHub Actions, React/TypeScript
• Designed multi-step Claude API prompt chains that extract structured JSON from resumes and job postings and generate weighted, explainable compatibility scores, with backend validation to reject malformed AI responses before they reach users.
• Built FastAPI REST endpoints, PostgreSQL models, SQLAlchemy ORM layers, Alembic migrations, and a React/TypeScript frontend; developed an Airflow ingestion pipeline for repeatable job-data updates.
• Authored the platform's full AWS deployment in Terraform (ECS Fargate, ALB, RDS PostgreSQL, VPC, IAM, CloudWatch) behind a credential-free CI pipeline running a mocked-provider test suite, tfint, and Trivy security scanning on every pull request.
• Hardened the configuration against 5 HIGH/CRITICAL static-analysis findings and codified production decisions as regression tests, including separating liveness from readiness probes after finding the readiness check shared a connection pool with application traffic.

SurSadhana AI | Vocal Pitch Trainer for Indian Classical Music
React, FastAPI, NumPy, Web Audio API
• Built a full-stack singing practice app that records vocals, extracts pitch contours with a custom YIN-style detector, scores them against harmonium/sargam targets in cents, and classifies vocal stability from signal-processing features.
• Implemented a real-time tuner by porting the pitch detector into an AudioWorklet running on the browser's audio thread, giving live cents-offset feedback without a server round-trip.
• Vectorized backend pitch detection with NumPy using an FFT-based difference function, cutting analysis latency 59x (717ms to 12ms) at identical accuracy, 100% note detection and 1.83-cent mean error on a 96-recording synthetic benchmark.
• Designed onset-aware sequence scoring with a monotonic Viterbi alignment mapping sung frames to target notes at actual pitch changes, modeled all ten Hindustani thaats, and shipped as an installable PWA with 82 automated tests (45 backend, 37 frontend) in CI.

Alberta Energy Data Pipeline | ETL & Analytics System

Python, Apache Airflow, PostgreSQL, dbt, Docker, GitHub Actions, Power BI

- Built an Airflow-orchestrated ETL pipeline ingesting Alberta Energy Regulator production data, with a validation and audit layer that quarantines invalid records (nulls, bad dates, negative volumes, duplicates) into dedicated quality-issue tables and tracks per-run row counts and error rates.
- Implemented idempotent incremental loading using chunked upserts against natural-key unique indexes, and modeled an analytics layer in dbt (staging views, a unified fact table, summary marts) enforced by dbt data tests and consumed by Power BI.
- Dockerized the full stack and built a two-stage GitHub Actions CI pipeline: unit tests plus an end-to-end integration job that runs the pipeline twice against live PostgreSQL to prove upsert idempotency, then executes dbt build.

UnderdogManager | C++ Simulation Engine (Unreal Engine 5.8) github.com/jaskeeratr/underdog-manager

C++, Unreal Engine, Slate/UMG

- Architected a public C++-first basketball management simulation with module boundaries, deterministic league systems, and zero external asset dependencies, across an 11,000+ line codebase with 21 static service classes and a 22-screen Slate/UMG dashboard.
- Designed a generated 12-team/180-player league with a validated 22-game double round-robin schedule and a deterministic possession-based match simulator covering overtime, standings, playoffs, player development, injuries, morale, contracts, free agency, draft, and AI management.
- Implemented async save snapshots with versioned schema validation and deterministic random number generation, and wrote 47 passing automated tests including soak, schedule integrity, match integrity, roster, staff-payroll, and franchise-cash tests.

Multi-Sport Match Outcome Predictor

Python, Flask, SQLite, scikit-learn

- Developed a full-stack ML platform predicting match outcomes across 5 sports using engineered features, adaptive retraining workflows, injury adjustments, and confidence scoring.
- Achieved 66.8% accuracy across 1,140+ Premier League matches, outperforming a naive baseline by 17%; built Flask REST APIs, SQLite persistence, prediction history, and what-if simulations while improving SQL feature extraction 22% through profiling.

Cinematic Developer Portfolio

jaskeerat-portfolio.vercel.app

React, TypeScript, Three.js, GSAP

- Designed and shipped a React/TypeScript portfolio with Three.js/React Three Fiber 3D scenes, scroll-driven storytelling via GSAP and Framer Motion, and performance-conscious mobile fallbacks.

LEADERSHIP

Co-President | Sikh Student Association, University of Calgary

2024 -- Present

- Led a 100+ member student organization across cross-functional executive teams, planning and executing 10+ events annually with budget, timeline, and stakeholder ownership.
- Coordinated communication between student executives, university administration, and community partners, adapting communication style for technical and non-technical audiences.